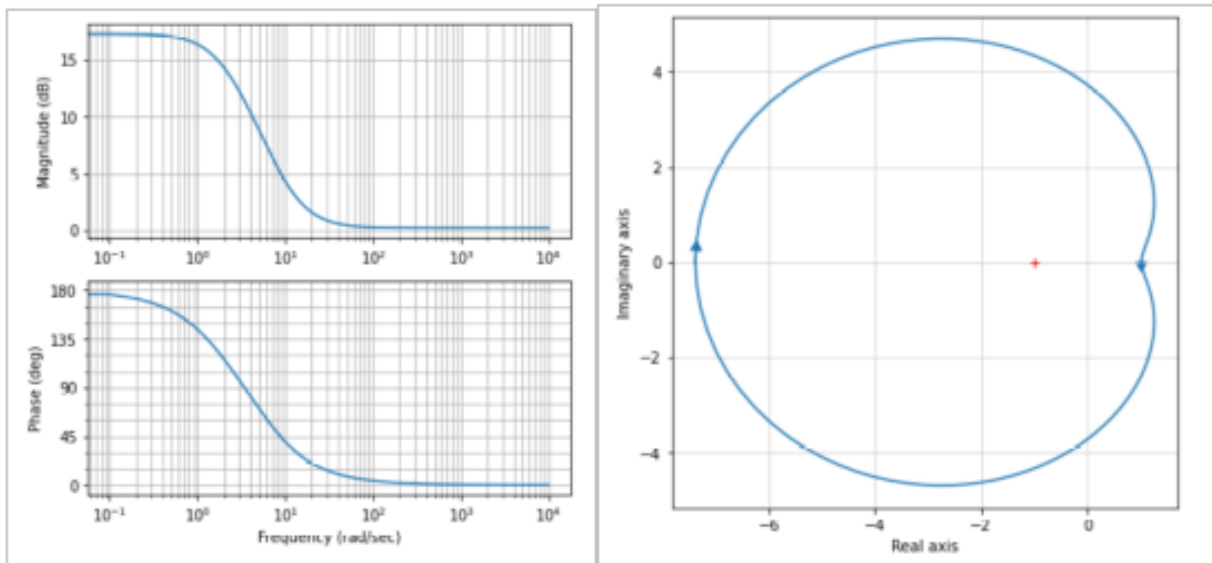


University of Cape Town
Department of Electrical Engineering
EEE3094S - Control Systems Engineering
Problem Set 5.1: Nyquist Plots

Question 1

A given system is unstable in closed-loop.

- A) Explain how you would know that from looking at each of the frequency domain plots below
- Bode plot
 - Nyquist plot
- B) How many closed-loop poles are in the right-half plane?
- C) Approximate how much you would have to reduce the gain by to stabilise the system.



Solution:

A) i. Bode plot: The apparent lack of gain and phase margins indicates that the system may be unstable (the magnitude plot never reaches 0 dB while the phase plot never reaches 180°). Stability is easier to analyse with other methods.

ii. Nyquist plot: The Nyquist plot encircles -1 in a clockwise direction. There must be at least one unstable pole in the system.

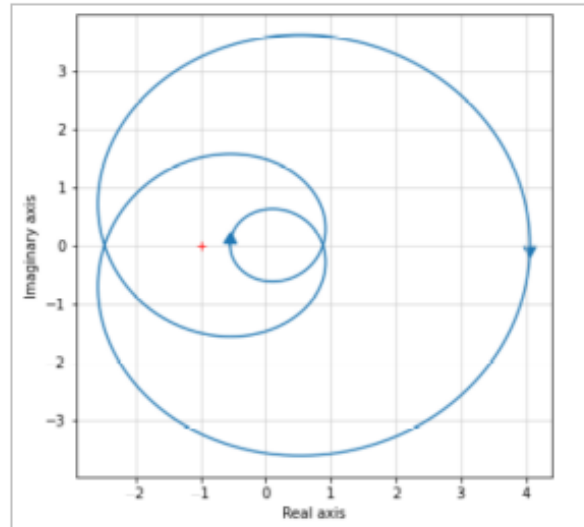
B) One, due to the single encirclement in the Nyquist plot.

C) The system crosses through approximately -7 in the Nyquist plot. The gain would have to be reduced by a factor of $1/7$ to stabilise the system.

Question 2

The following Nyquist plot describes a third-order system that is stable in open-loop.

- A) How many unstable closed-loop poles does it have at its present gain?
- B) Will the number of unstable closed-loop poles change if the gain is increased?
- C) How many right-half plane zeros do you expect this system to have based on your previous answers?



Solution:

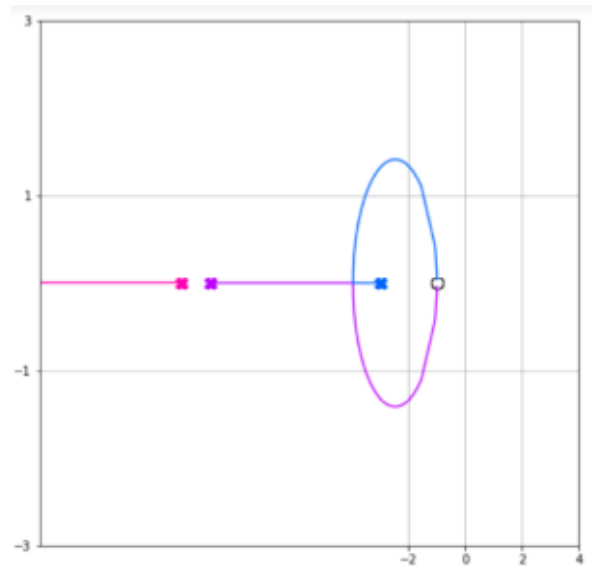
A) There is one clockwise encirclement of -1 in the Nyquist plot. This means there is at least 1 unstable pole in the RHP.

B) Yes, if the gain is increased, the smaller clockwise circle around the origin will expand to encircle -1, which indicates that an additional pole will become unstable.

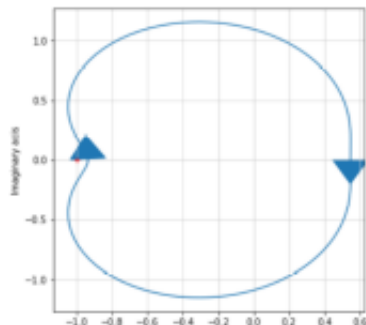
C) The number of clockwise encirclements of -1 tell us N , where $N = Z - P$. Given that the system is stable in open loop, there should be 2 RHP zeros because there are up to 2 encirclements in the system.

Question 3

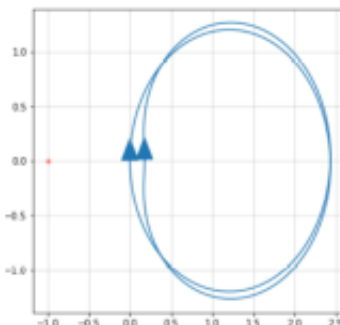
The system associated with the following root locus is closed-loop stable at its present gain.



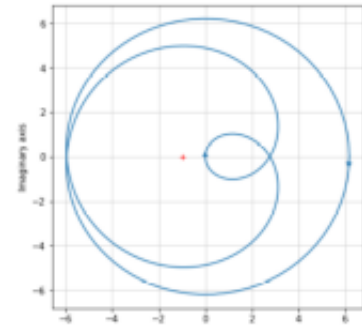
Which of the following Nyquist plots would you say belongs to this system? If a plot does not apply, give a brief reason that you are excluding it.



A



B



C

Solution:

A: As the gain of A is increased, it will become unstable with at least 1 pole in the RHP. The root locus above will not become unstable at high gain due to the zeros being in the LHP. Therefore, it does not belong with the system.

B: This one belongs with the system. The plot does not cross the y-axis. This will make sure the system remains stable with increasing gain.

C: The system circles -1 twice. This indicates there are at least 2 poles in the RHP. This is not supported by the root locus plot, and therefore, C does not represent the system.